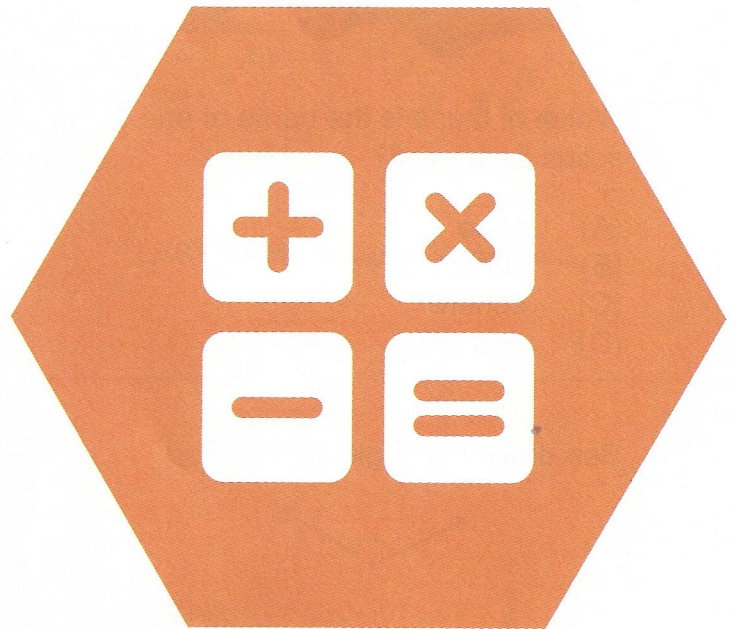




UNSW Global
AUSTRALIA

PAPER B



2016 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

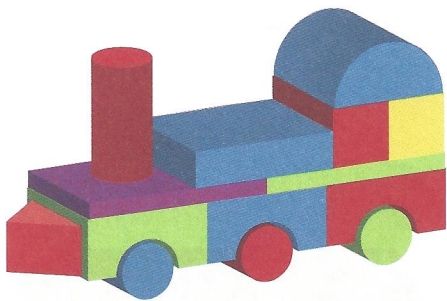
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. Sam built a train using 14 solids.



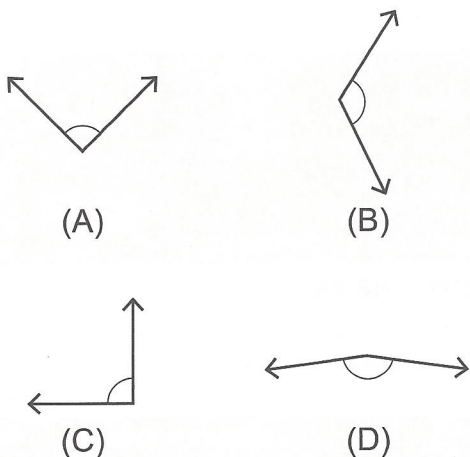
Which of these is the name of one of the solids Sam used?

- (A) cone
- (B) cube
- (C) sphere
- (D) cylinder

2. Ben drew this angle.



Which angle is the same size as Ben's?



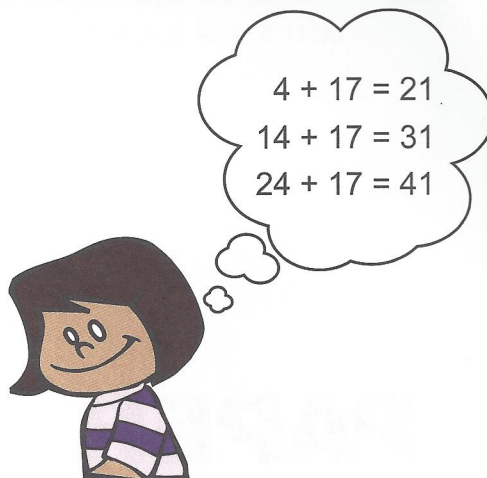
3. Jo wrote this number pattern.

1, 5, 9, 13, ?, 21, 25, ...

What number must ? be?

- (A) 14
- (B) 16
- (C) 17
- (D) 20

4. Elyse was practising addition.



$44 + 17 =$?

What must the value of ? be?

- (A) 51
- (B) 61
- (C) 71
- (D) 81

5. Which container will hold about one cup of liquid when full?



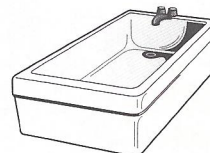
(A)



(B)

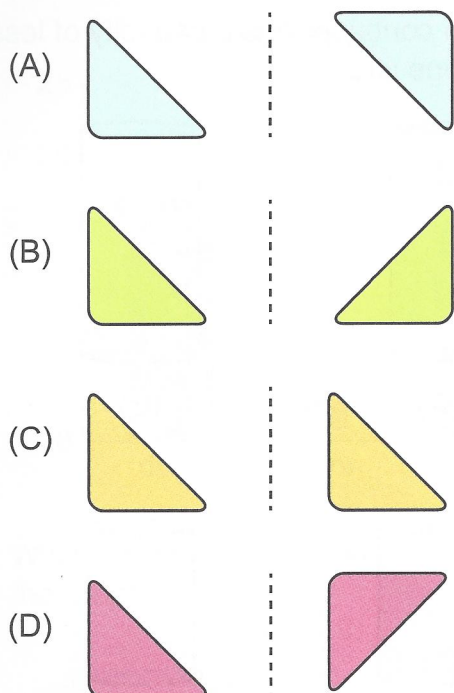


(C)

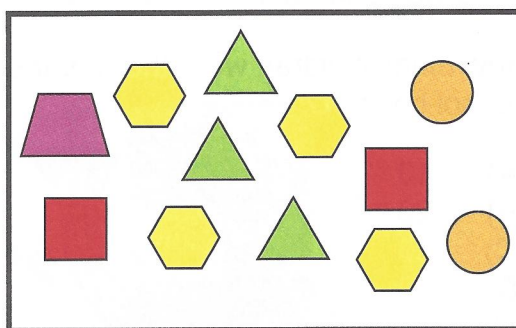


(D)

6. Which of these shows a shape reflected in the dotted line?



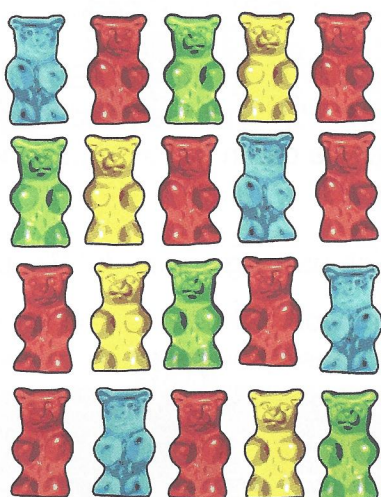
8. Tom has 12 plastic shapes. He closes his eyes and picks up a shape.



Which shape has twice the chance of being picked up as the square?

- (A) (B)
- (C) (D)

7. Ella had 20 jelly bears.

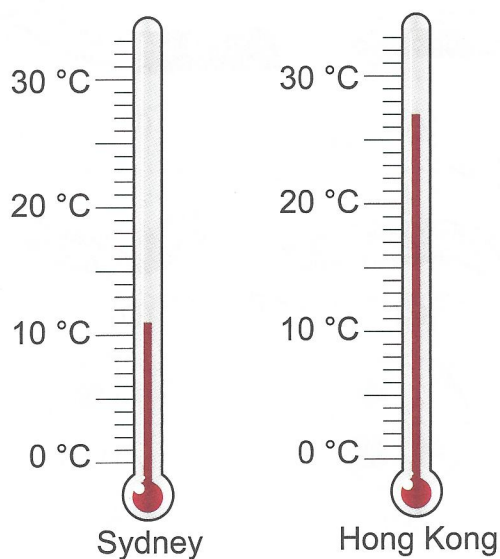


She had 6 bags. Ella placed the same number of jelly bears in each bag.

How many jelly bears did Ella have left over?

- (A) 1
(B) 2
(C) 3
(D) 4

9. These thermometers show the temperatures in two cities one morning.



How much warmer was it in Hong Kong than in Sydney?

- (A) 13 °C
(B) 15 °C
(C) 16 °C
(D) 18 °C

10. There were 60 people altogether at the school concert. For every two children there was one adult.

How many children were there at the school concert?

- (A) 10
- (B) 20
- (C) 30
- (D) 40

11. Kim is putting a train track together.



Which two pieces must she use to complete the track?

- 

(A)
- 

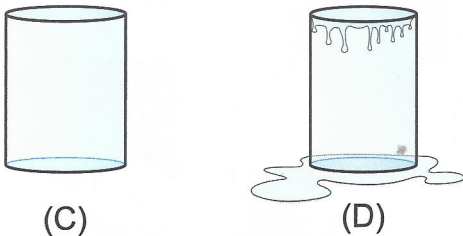
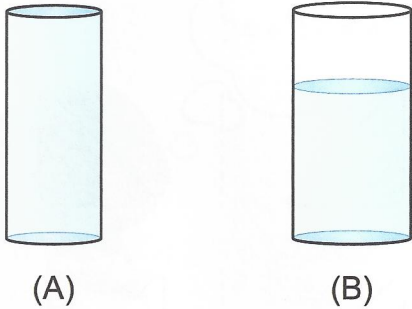
(B)
- 

(C)
- 

(D)

12. Sue poured one litre of water into each of the containers shown.

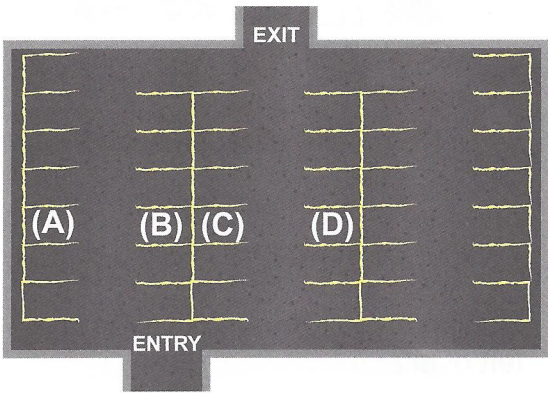
Which container has a capacity of less than one litre?



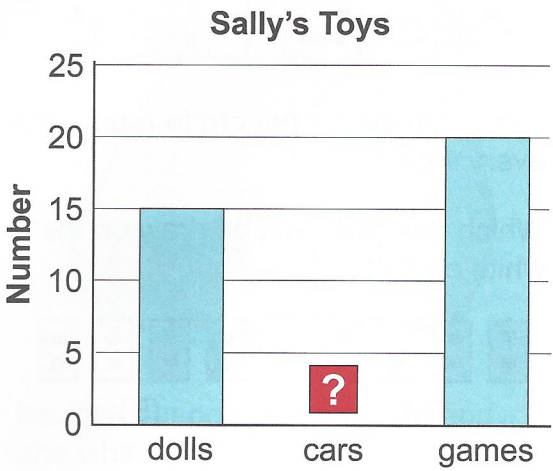
13. This is a map of a car park.

Raj drove his car into the ENTRY of the car park. He turned left and then right. He parked his car in the third parking space on his left.

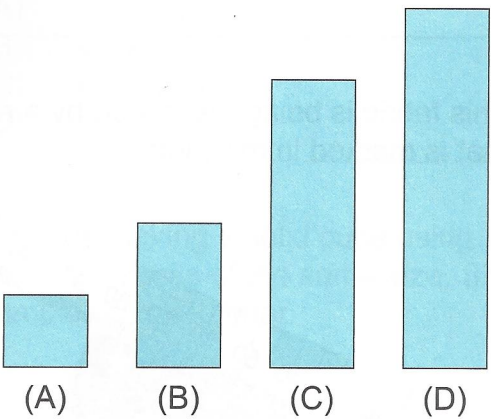
In which parking space did Raj park his car?



14. Sally organised 45 of her toys into three groups.



Which of these should replace **?** on the graph?



15. Jane asked her friends, 'How much pocket money do you get each week?'

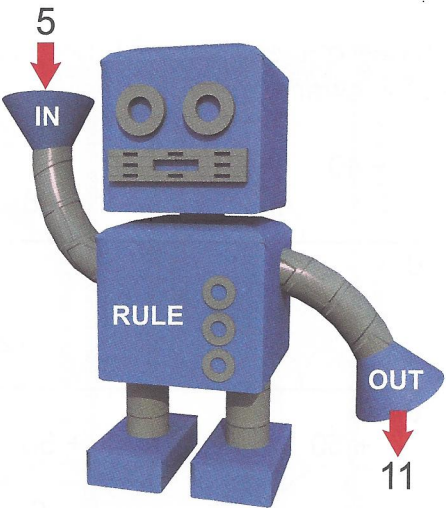
After they had all answered, Chris said, 'I get less than Abby but \$3 more than Deb.'

Billy said, 'I get \$1 less than Chris.'

Who gets the most pocket money?

- (A) Abby
(B) Billy
(C) Chris
(D) Deb

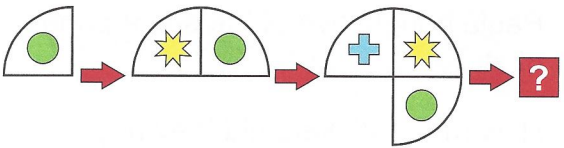
16. John has a robot. He put in the number 5. The robot applied a rule then gave out the number 11.



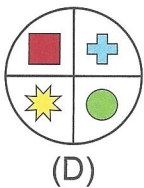
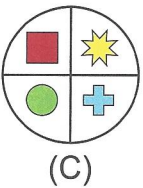
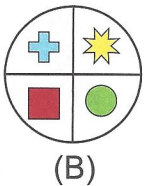
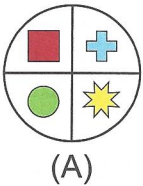
What could the robot's rule be?

- (A) subtract 1 from the number then halve it
(B) double the number then add 1
(C) triple the number then subtract 3
(D) add 7 to the number

17. Jack follows this pattern to form a shape.



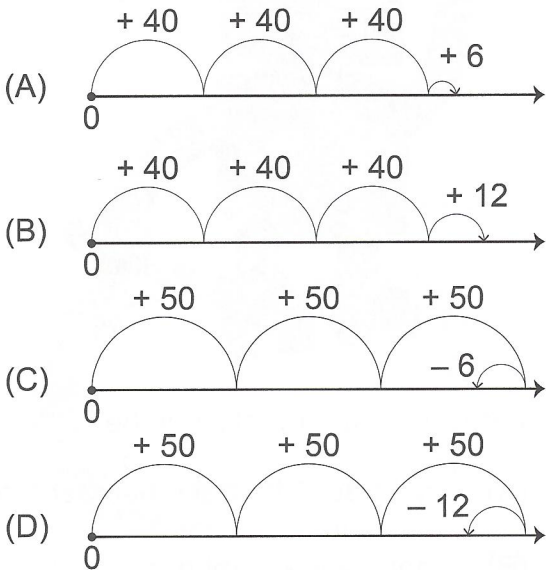
Which of these is his final shape?



18. Maddie used a number line to solve this number sentence.

$$46 \times 3 = \boxed{?}$$

Which one of these shows how to find the correct answer?



19. Janine bought three pages of stickers with 30 stickers on each page.

Paula bought five of the same pages of stickers.

How many stickers did they buy altogether?

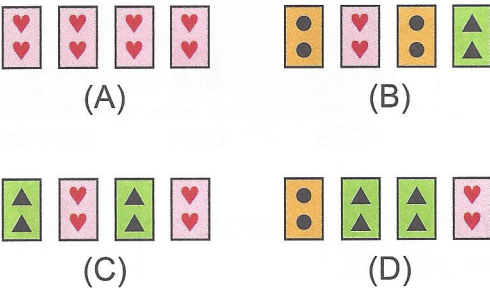
- (A) 38
- (B) 95
- (C) 240
- (D) 450

20. Indra had 12 white cards. He drew designs on 8 of them.



Indra wanted the pattern to repeat after every 4 cards.

Which designs must he draw on the white cards?



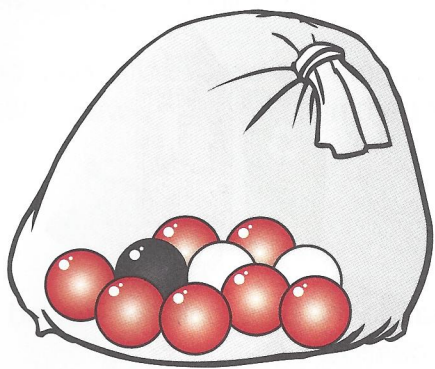
21. This fabric is being measured by a ruler that is marked in millimetres.



Which of these is closest to the length of the fabric?

- (A) 70 millimetres
- (B) 75 millimetres
- (C) 82 millimetres
- (D) 182 millimetres

22. Chandra had 7 red, 1 black and 2 white marbles in a bag.

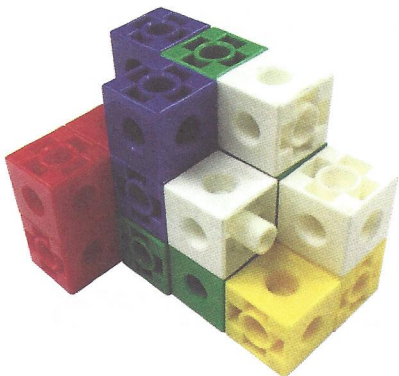


He took out one black marble and one white marble.

What is the chance that the next marble Chandra took out was black?

- (A) impossible
- (B) unlikely
- (C) even chance
- (D) likely

23. Sarah is making a solid cube using coloured blocks of the same size. She has completed one corner.

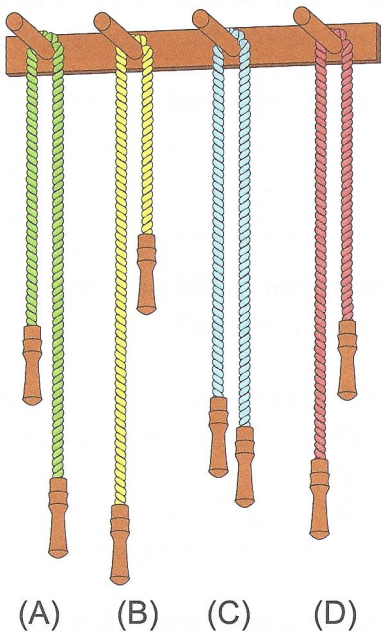


How many blocks has Sarah used so far?

- (A) 15
- (B) 19
- (C) 21
- (D) 23

24. There are four skipping ropes hanging on hooks in the school hall.

Which skipping rope is the shortest?



25. In Anish’s code, the symbols are added together to give values. The table shows the value of Anish’s sums using his symbols.

Anish’s symbols	Value
 	2
 	6
 	30
    	

What value must  be?

- (A) 38
- (B) 41
- (C) 42
- (D) 48

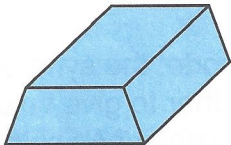
26. Aditi is a musician. She recorded the number of hours she spent practising for each concert over 4 weeks.

Week	Number of concerts	Hours of practice for each concert
1	2	4
2	3	2
3	1	5
4	2	4

How many hours did Aditi spend practising over these 4 weeks?

- (A) 15
- (B) 23
- (C) 27
- (D) 29

27. Ali made this solid using 6 pieces of cardboard.



Which one of these shows the pieces of cardboard Ali used?

- (A)
- (B)
- (C)
- (D)

28. This picture shows two bins for waste when viewed from the south.

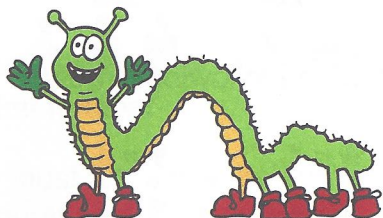


What do the bins look like viewed from the east?

- (A)
- (B)
- (C)
- (D)

29. A caterpillar crawls 1 metre on the first day, 2 metres on the second day, and then 3 metres on the third day.

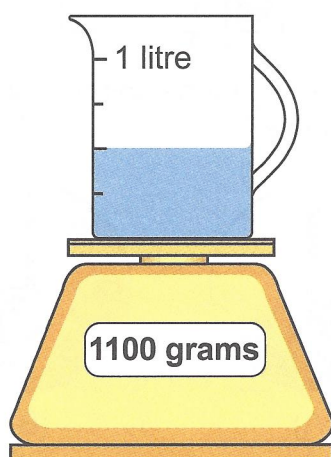
The caterpillar keeps crawling in this way until it crawls 10 metres on the tenth day.



How many metres in total will the caterpillar crawl in ten days?

- (A) 66
- (B) 55
- (C) 45
- (D) 10

30. Ying measured the mass of a jug containing some water.



One litre of water has a mass of 1 kilogram.

What was the mass of the jug?

- (A) 100 grams
- (B) 500 grams
- (C) 550 grams
- (D) 600 grams

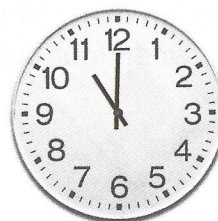
31. In Mrs Li's class today each student did either reading or writing.

- There were groups of 3 for reading.
- There were groups of 5 for writing.
- There were 7 groups altogether.

How many students could be in the class?

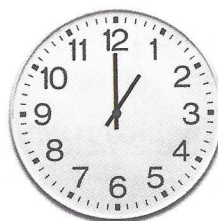
- (A) 21
- (B) 24
- (C) 26
- (D) 29

32. Sven had to deliver 200 leaflets. He looked at the clock as he left home.

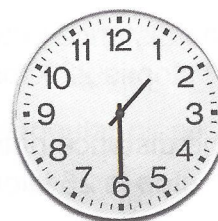


Sven delivered 20 leaflets every 15 minutes.

Which clock shows the time Sven finished his delivery?



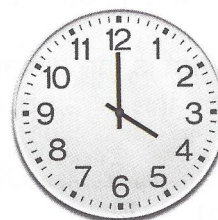
(A)



(B)



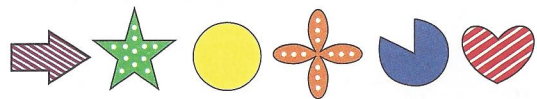
(C)



(D)

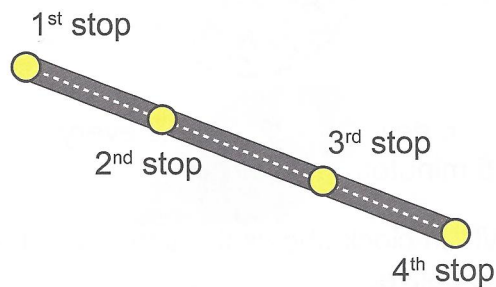
33. Lori said, "A striped shape will have only curved edges. A plain shape will have only straight edges. A spotted shape will have both curved and straight edges."

How many of these shapes fit Lori's description?



- (A) 1
- (B) 2
- (C) 3
- (D) 4

34. Lin drew a rough map to help her find the distances between four bus stops along a country road.



The distance between the 1st stop and the 3rd stop is 22 kilometres.

The distance between the 2nd stop and the 4th stop is 25 kilometres.

The distance between the 1st stop and the 4th stop is 33 kilometres.

What is the distance between the 2nd stop and the 3rd stop, in kilometres?

- (A) 8
- (B) 11
- (C) 14
- (D) 17

35. These calculations give the same answer. The answer is greater than 60.

27
+ ☆ 4

?

83
- ♥ 2

?

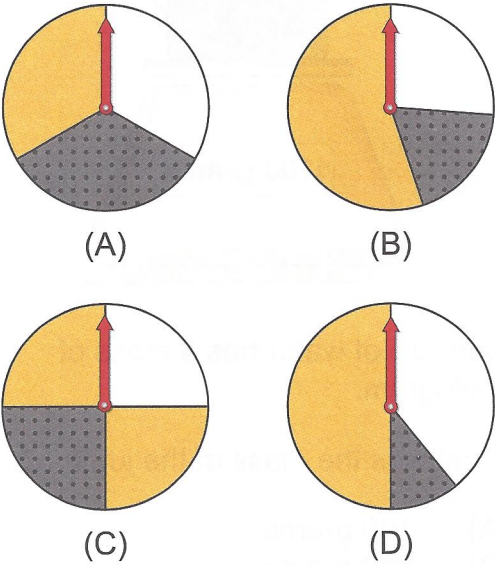
Which of the following is true?

- (A) ☆ = 1 ♥ = 4
- (B) ☆ = 2 ♥ = 3
- (C) ☆ = 3 ♥ = 2
- (D) ☆ = 5 ♥ = 1

36. Liam spun a spinner 18 times and recorded his results.

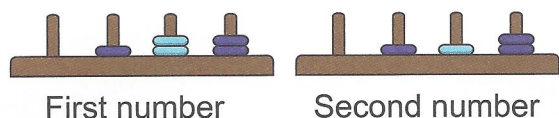
	###
	###

Which one of these spinners is most likely to be Liam's?

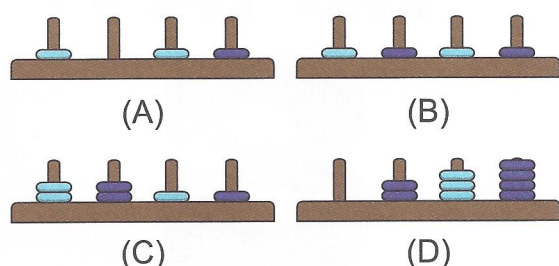


37. Raj invented a new number system for his abacus. When the number of rings on a post reaches 3, they are replaced by one ring on the next post to the left.

He made two numbers using his abacus.



How must Raj show the sum of these two numbers on his abacus?



38. Anna found a page from a 2008 calendar.

May 2008						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17

This note was on the page.



When did Adriana go to France?

- (A) Thu 22 May 2008
(B) Thu 29 May 2008
(C) Wed 4 June 2008
(D) Thu 5 June 2008

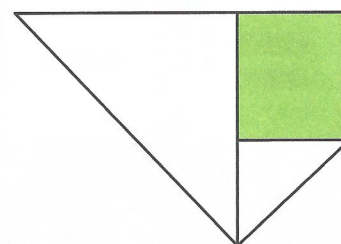
39. Nine identical squares are joined together to form a large square.

The area of the large square is 36 square centimetres.

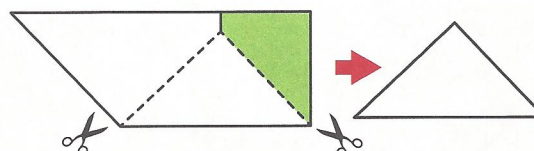
What is the distance around the outside of one small square?

- (A) 4 centimetres
(B) 6 centimetres
(C) 8 centimetres
(D) 16 centimetres

40. Amy had a green, rectangular piece of paper. She folded it twice, as shown.

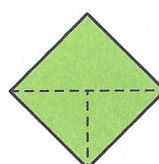


Amy then made another fold and cut out the triangular piece.

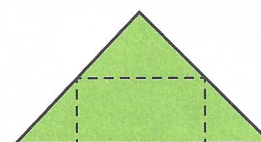


Amy unfolded this triangular piece of paper.

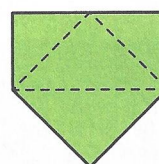
Which of these is the shape of Amy's unfolded paper?



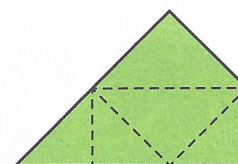
(A)



(B)



(C)



(D)

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Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific Region	Year 5
Singapore	Primary 3
South Africa	Grade 4



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TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: 4 + 6 =

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval (C), as shown.

(A) (B) (C) (D)



START

1 (A) (B) (C) (D)	11 (A) (B) (C) (D)	21 (A) (B) (C) (D)	31 (A) (B) (C) (D)
2 (A) (B) (C) (D)	12 (A) (B) (C) (D)	22 (A) (B) (C) (D)	32 (A) (B) (C) (D)
3 (A) (B) (C) (D)	13 (A) (B) (C) (D)	23 (A) (B) (C) (D)	33 (A) (B) (C) (D)
4 (A) (B) (C) (D)	14 (A) (B) (C) (D)	24 (A) (B) (C) (D)	34 (A) (B) (C) (D)
5 (A) (B) (C) (D)	15 (A) (B) (C) (D)	25 (A) (B) (C) (D)	35 (A) (B) (C) (D)
6 (A) (B) (C) (D)	16 (A) (B) (C) (D)	26 (A) (B) (C) (D)	36 (A) (B) (C) (D)
7 (A) (B) (C) (D)	17 (A) (B) (C) (D)	27 (A) (B) (C) (D)	37 (A) (B) (C) (D)
8 (A) (B) (C) (D)	18 (A) (B) (C) (D)	28 (A) (B) (C) (D)	38 (A) (B) (C) (D)
9 (A) (B) (C) (D)	19 (A) (B) (C) (D)	29 (A) (B) (C) (D)	39 (A) (B) (C) (D)
10 (A) (B) (C) (D)	20 (A) (B) (C) (D)	30 (A) (B) (C) (D)	40 (A) (B) (C) (D)

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PAPER

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2016 ICAS

International Competitions and Assessments for Schools

PAPER B – MATHS 2016, ICAS ANSWER KEY

QUESTION NUMBER	PAPER B
1	D
2	B
3	C
4	B
5	C
6	B
7	B
8	A
9	C
10	D
11	C
12	D
13	A
14	B
15	A
16	B
17	A
18	D
19	C
20	C
21	B
22	A
23	C
24	B
25	B
26	C
27	D
28	A
29	B
30	D
31	D
32	B
33	A
34	C
35	C
36	D
37	A
38	D
39	C
40	D